

AHAMED SAFNAS

Sri Lanka

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EXECUTIVE SUMMARY

Associate Software Engineer (AI/ML) skilled in Python, SQL and data analysis, with experience cleaning, validating and transforming large datasets. Developed end-to-end MLOps pipelines using Docker, Kubernetes and DVC, and built retrieval systems with LLMs for secure data insights. Holds Data Scientist and Data Analyst certifications and strong communication skills, seeking a Junior Data Scientist role.

PROJECTS

MultiDocChat : Production-Ready Conversational RAG System [MultiDocChat : Production-Ready Conversational RAG System](#)

- Built a multi-document chatbot using FastAPI, LangChain, and FAISS that allows users to upload PDF, DOCX, and TXT files and chat with their content using Groq and Gemini LLMs with session-based memory.
- Implemented MMR-based retrieval with HuggingFace embeddings and an LCEL pipeline that rewrites follow-up questions as standalone queries for accurate multi-turn conversations.
- Set up a LangSmith evaluation pipeline using a custom LLM-as-a-Judge evaluator to measure answer correctness against a golden dataset and visualized performance trends using Tableau dashboards
- Automated CI/CD using GitHub Actions to run tests, build a Docker image, push it to AWS ECR, and deploy to ECS Fargate with secrets managed through AWS Secrets Manager.

VisaFlow : A Production-Ready MLOps Pipeline for Visa Approval Prediction

[VisaFlow : A Production-Ready MLOps Pipeline for Visa Approval Prediction](#)

- Architected and deployed a production-ready ML pipeline in Python and SQL for US visa approval prediction, automating the lifecycle from data ingestion to model serving
- Optimized data scalability by integrating AWS S3 and MongoDB, reducing data retrieval latency and enabling robust model versioning for team collaboration.
- Containerized the ecosystem using Docker and developed RESTful APIs (FastAPI), ensuring 100% environment reproducibility across dev, staging, and production.
- Engineered automated MLOps workflows with GitHub Actions and YAML-based configurations, achieving seamless CI/CD and continuous model retraining.
- Performed data cleaning, transformation, and validation using Pandas and SQL, and applied statistical analysis to support model feature engineering
- Created interactive Power BI dashboards to present prediction accuracy and business insights to stakeholders

Medical Chatbot [Medical Chatbot](#)

- Developed an end-to-end medical chatbot using Google Gemini API, LangChain, and Pinecone, enabling context-aware, real-time medical query responses through a Flask web application.
- Built a Retrieval-Augmented Generation (RAG) pipeline to process medical PDFs, generate vector embeddings, and perform efficient semantic search with Pinecone.
- Designed and managed both backend (Python, Flask, LangChain) and frontend (HTML, CSS, Bootstrap) components, ensuring secure environment configurations and optimized performance.
- Designed and deployed a CI/CD pipeline using GitHub Actions and AWS (EC2, S3, IAM) for automated testing and deployment.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, TypeScript, R
- **Data Engineering & Large-Scale Processing:** PySpark, Distributed Data Processing, ETL Pipelines, Data Transformation, Data Ingestion Pipelines, Batch Processing, Data Validation
- **Generative AI & LLM Engineering:** LangChain, Hugging Face Transformers, OpenAI API, Gemini API, Llama 2/3, Mistral, Prompt Engineering, Fine-Tuning (QLoRA), AI Agents, Agentic Workflows, Multi-Agent Systems
- **Retrieval & Knowledge Systems:** RAG, Semantic Search, Embedding Models, Pinecone, ChromaDB, FAISS, pgvector, Hybrid Search, Metadata Filtering, MMR Retrieval, Re-Ranking (Cross-Encoder/BGE Re-Ranker), GraphRAG, Vectorless Retrieval Architectures
- **Machine Learning, Statistics & Data Analysis:** TensorFlow, Keras, Pandas, Model Training, Hyperparameter Tuning, Model Evaluation, Statistics, Data Analysis
- **MLOps & AI Platform Engineering:** Docker, Kubernetes, DVC, MLflow, Git, GitHub Actions, CI/CD Pipelines, Model Versioning, Experiment Tracking, Automated Deployment, Monitoring & Observability
- **Cloud & Infrastructure:** AWS (EC2, ECS, ECR, S3, IAM, Secrets Manager), Google Cloud Platform (GCP), Linux
- **Backend & System Design:** FastAPI, Flask, REST APIs, Microservices Architecture, Distributed Systems, System Design, Scalable AI Architecture, API Design

- **AI Security & Responsible AI:** IAM & Role-Based Access Control (RBAC), Secrets Management, Secure Model Deployment, Data Privacy, Secure Prompting, LLM Guardrails
- **Databases & Data Systems:** PostgreSQL, MySQL, MongoDB
- **Frontend:** React.js, Next.js, Node.js
- **Data Visualization & Reporting:** Data Visualization, Power BI, Tableau
- **Soft Skills:** Problem Solving, Team Collaboration, Analytical thinking, Communication, Rapid Learning, Critical Analysis, Adaptability

WORK EXPERIENCE

AI THARIO

Jan 2025 - Mar 2026

Associate Software Engineer(AI/ML)

Dubai, UAE(Remote)

- Designed and deployed end-to-end MLOps pipelines using DVC, Docker, and Kubernetes, automating model training, versioning, production deployment, and data ingestion pipelines with CI/CD workflows ensuring zero-downtime updates
- Implemented experiment tracking using MLflow, enabling systematic comparison of model versions, hyperparameters, and evaluation metrics across teams
- Developed and deployed production-grade RAG pipelines integrating Llama 2/3 and Mistral with Pinecone and ChromaDB vector stores via LangChain, enabling businesses to query private data securely
- Built and shipped AI chatbot assistants leveraging fine-tuned LLMs and advanced prompt engineering techniques to mitigate hallucinations and improve contextual response accuracy
- Fine-tuned Llama 2/3 and Mistral models using via HuggingFace for domain-specific tasks, maintaining low computational footprint without sacrificing performance
- Analyzed structured and unstructured datasets using Python and R, performing statistical analysis to uncover trends and support business decision-making
- Created interactive dashboards in Tableau and Power BI to visualize key metrics and insights for stakeholders

EDUCATION

NED University

2021 - 2025

B.E., Computer & Information System Engineering

- **GPA:** 3.7/4.0
- **Achievements:** 1st Class

CERTIFICATES

- AI Fundamentals
- Data Scientist Associate
- Data Analyst Associate
- Cloud Computing Fundamentals

LANGUAGES

- English
- Urdu
- Hindi
- Tamil
- Sinhala